

RayLab turns idle lab machines into one shared compute pool.

A cross-platform RayLab cluster ran a Monte Carlo option-pricing simulation across Windows and macOS nodes. The same 480 million simulated paths finished more than twice as fast when a second machine joined the job.

Measured cluster run

MEASURED SPEEDUP
2.26x
Two nodes vs. one node on the same Monte Carlo workload.

BEST WALL TIME
3.44s
Down from 7.77s with one node.

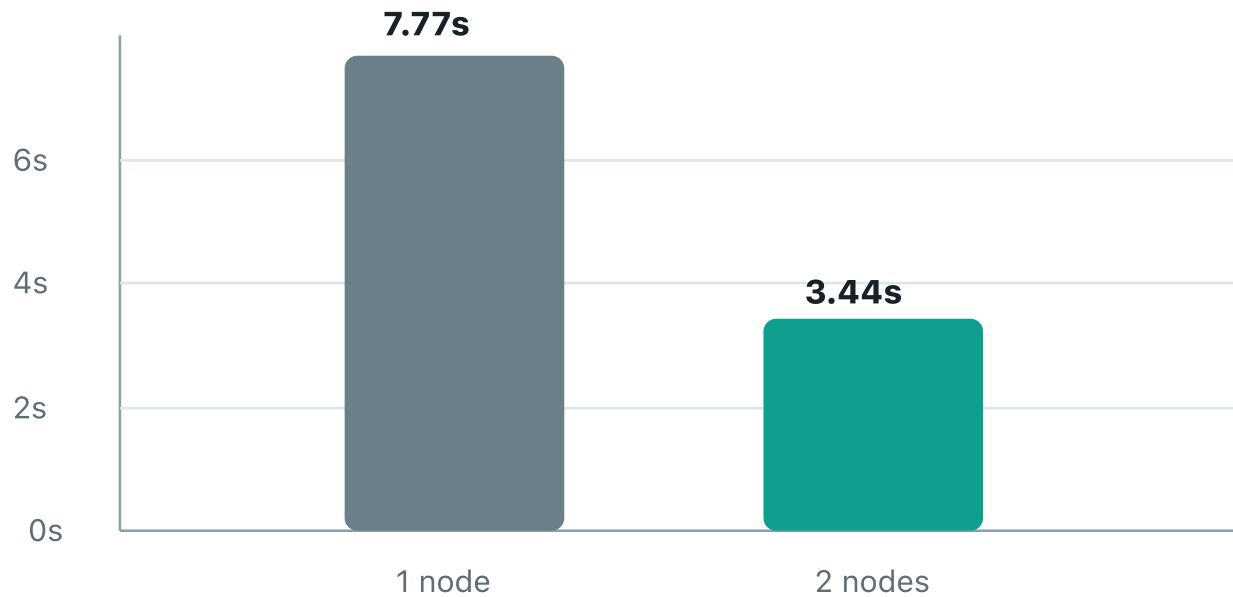
WORKLOAD
480M
Independent simulated market paths per run.

VALIDATION
0.0009
Monte Carlo standard error against Black-Scholes.

Performance Results

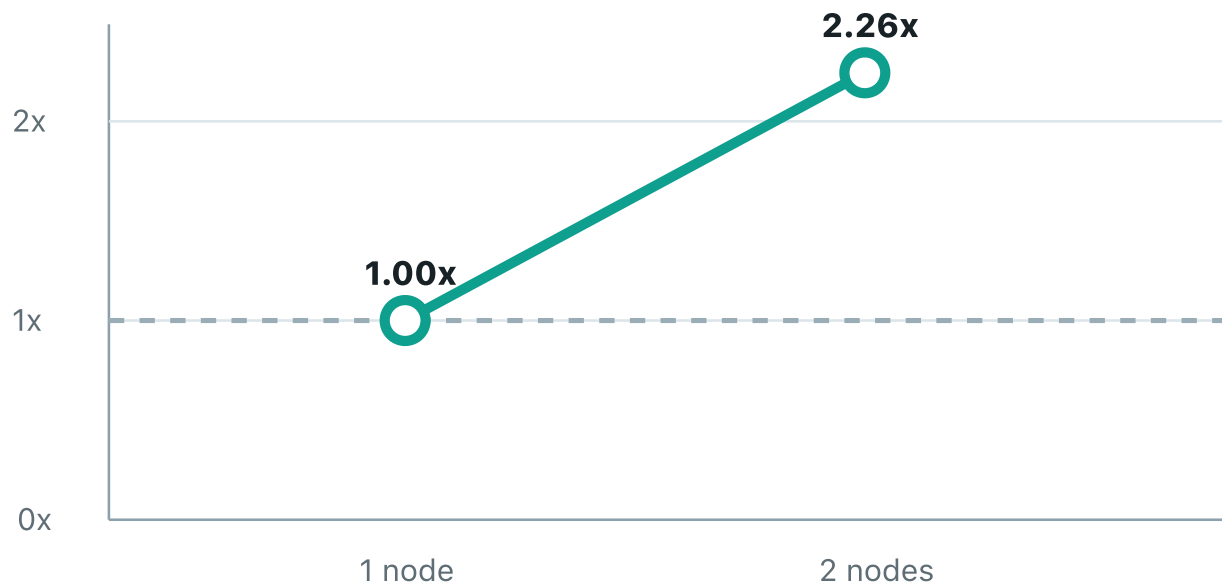
Task: price a European call option with 96 independent Monte Carlo chunks of 5,000,000 paths each.

Wall Time
Lower is better. Adding one worker cut the simulation runtime by more than half.



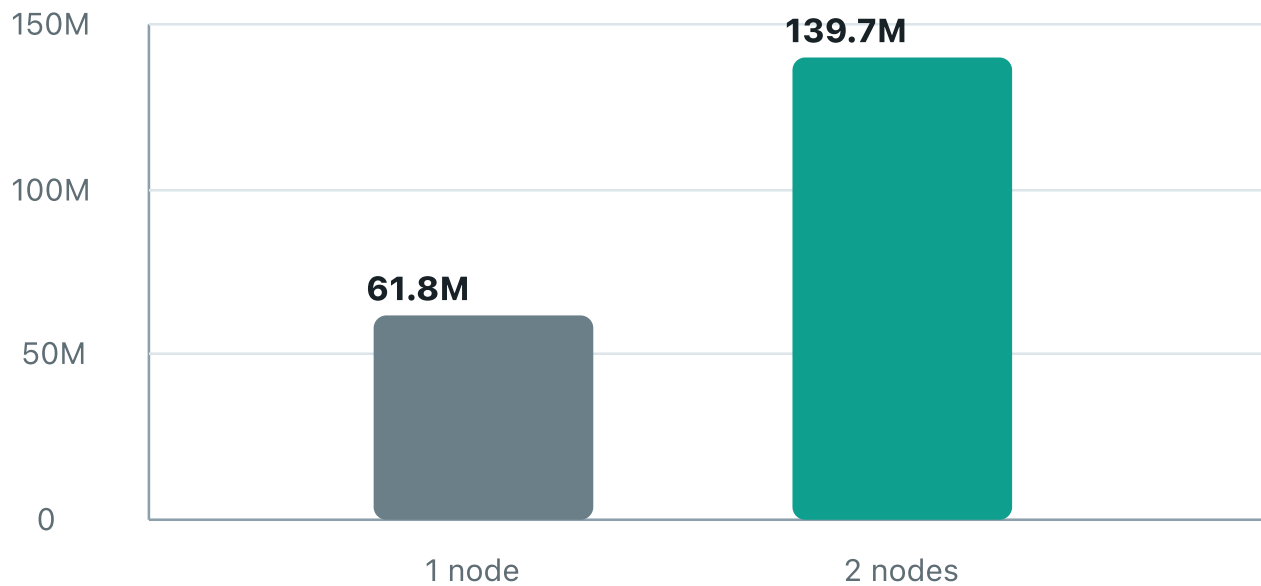
Speedup vs. One Node

Higher is better. This workload scales cleanly because every simulation chunk is independent.



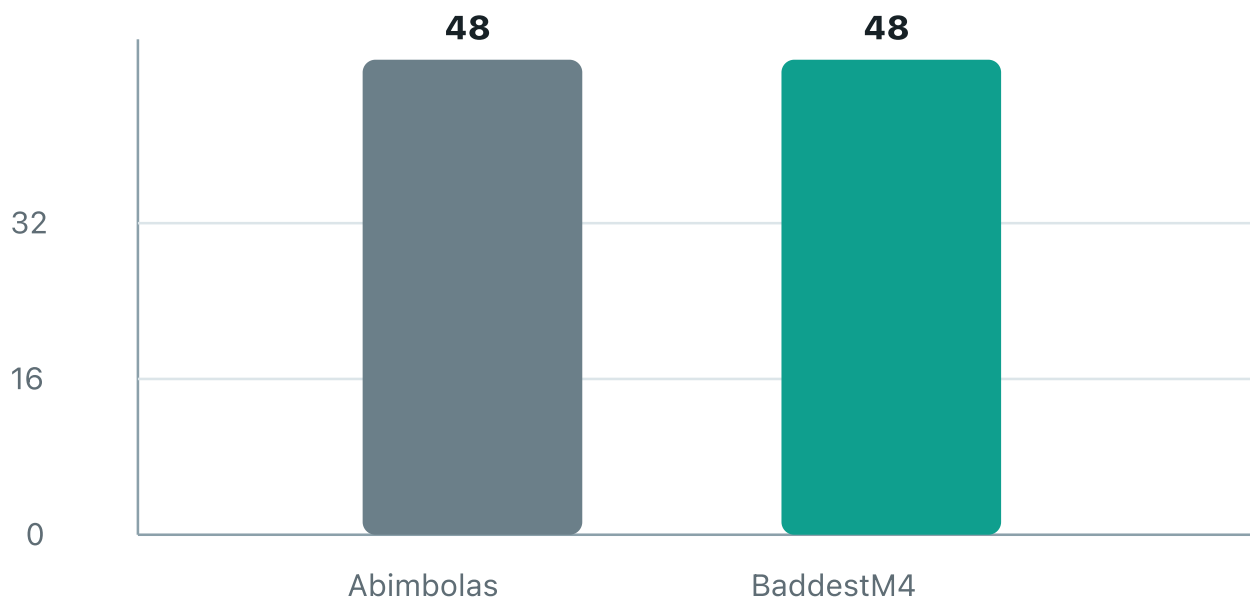
Throughput

Simulated paths completed per second. RayLab more than doubled useful throughput.



2-Node Work Distribution

Each node received 48 chunks. This confirms the simulation used both participating machines.



Contributing Nodes

One Windows coordinator and one macOS worker ran the clean demo workload.

Abimbolas

Windows coordinator at 192.168.33.12

Chunks	48
Paths	240M

BaddestM4.local	
macOS worker at 192.168.33.14	
Chunks	48
Paths	240M

What This Demonstrates

- RayLab can run one Python simulation over a mixed Windows and macOS cluster.
- The same 480 million paths finished 2.26x faster when RayLab used two nodes instead of one.
- Monte Carlo is a strong demo workload because every chunk is independent and has tiny result data.
- The computed option price matched the analytical Black-Scholes price within the measured standard error.

Benchmark Data			
NODES	SECONDS	PATHS/S	SPEEDUP
1	7.77	61.8M	1.00x
2	3.44	139.7M	2.26x
RESULT			VALUE
Black-Scholes price			11.539831
Monte Carlo price			11.539097
Standard error			0.000945

Measured on RayLab against coordinator 192.168.33.12. Task: European call option pricing using 96 independent Monte Carlo chunks of 5,000,000 simulated paths each. Hardware, network conditions, and workload shape affect scaling; this

report is intended as an honest proof that RayLab pooled real cross-platform compute resources.